

A Multi-modal BCI System Based on Motor Imagery

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Abstract—Brain-computer interface based on motor imagery is currently considered as the most promising brain-computer interface (BCI). By off-line analytic comparison, this paper used wavelet packet decomposition (WPD) and short-time Fourier transform (STFT) to feature extraction, which provided the basis for real-time online BCI system. With combination of advantages of two electroencephalograms (EEG) as Alpha wave and motor imagery and design of control strategy, multimodal brain-computer interface was built on LabVIEW to implement functions of mouse click and web browser. The experiment results suggested that the accuracy of four motor imagery movements were above 80% and the classification accuracy of research was at a ideal level. It proved that this system is feasible and has a high application value.

Keywords- motory magery ; Brain-computer interface; wavelet packet decomposition; LabVIEW

I. INTRODUCTION

Brain-computer interface (BCI) is a control system that realizes the communication and exchange between human and computer or other devices through EEG signals. It builds the direct communication and control channel between the human brain and other electronic devices by collecting and analyzing the bio-electroencephalogram of brain. And human can express or operate device through the brain instead of using language or body movement^[1-2].

Motor imagery refers particularly to the subject without any actual physical action, but just imagine some kind of action happening. When people think about limb movements (such as moving left and right hands), the brain's sensory-motor cortex exhibits rhythmic activity of the EEG. Motor imagery (MI) has arouse wide concern and become one of the current research hotspot because there are a great many potential application values in aspects like supplement of motor function and recovery of motor function, etc^[3]. Li Y combining Mu/Beta rhythm and P300 potential realise an EEG-based BCI system for 2-D cursor control^[4]. Kennedy's early intervention system successfully used in a patient for 17 months, the patient can always through the system gains the ability to control a computer mouse^[5]. However, electroencephalogram signals are nonstationary in long term measurement and extremely sensitive to noise. For these reasons, the extraction of ERD/ERS always suffers a low efficiency, poor stability and high dependence of user training. The users are not able to control the online system fully and automatically for it works under synchronous mode.

Therefore, this paper mainly searches for the optimal algorithm by carrying out off-line analysis and comparison for data to improve the accuracy and speed of online BCI system based on motor imagery, increase the control degrees of freedom and realize asynchronous work, and this system is feasible and has a high application value.

II. THE SYSTEM ARCHITECTURE

The collection section of system mainly consisted of ECG amplifier and electrode wire. It was supposed to be transmitted and saved in computer after completing the signal collection, amplification and A/D conversion. ECG amplifier took NT9200 while electrode wire used Ag-AgCl electrode, as shown in Fig. 1 and Fig. 2.



Figure 1 electrode wire



Figure 2 NT9200 amplifier

Firstly, it was to imagine the ECG upon these movements of left and right hand, feet and tongue. Secondly, ECG research

was supposed to be carried out from the following three aspects: preprocessing, feature extraction and model identification. ECG processing method that is suitable for four MI was selected by comparing classification results. Finally, it was to design control strategy in which alpha wave produced by short time eye-close achieved the asynchronous control for motor imagery and alpha wave produced by long time eye-close controlled to implement mouse click instruction to implement functions of mouse click and web browser based on the control that multimodal BCI to cursor.

LabVIEW is a graphical virtual instrument development platform launched by U.S. company^[7], The whole system was built on LabVIEW. It was shown in Fig. 3.

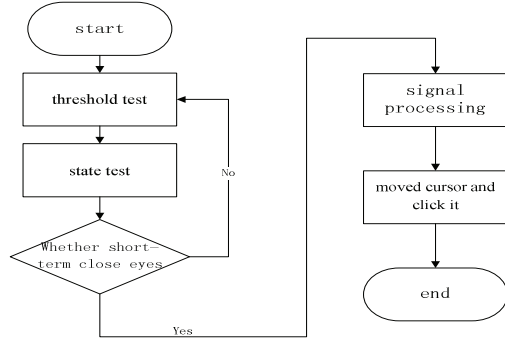


Figure 3 diagram programme of system

III. INTRODUCTION TO METHOD

A. Preprocessing

In preprocessing, FIR filter was used to filter for two signals to eliminate myoelectricity interference and power frequency interference. Independent component analysis was used to analyze the motor imagery EEG signals after filtering and Fast ICA based on maximum entropy was taken to remove eye artifacts. It was shown in Fig. 4.

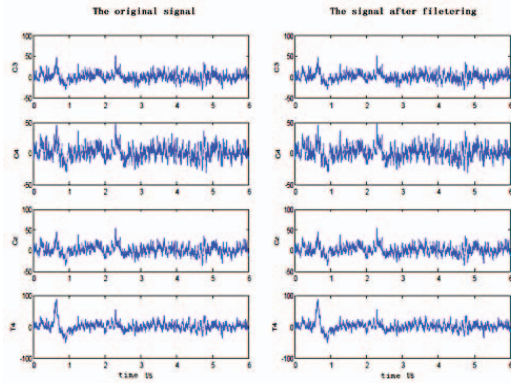


Figure 4 signal comparison between before and after filtering

As for measurement and test of threshold, the energy to alpha wave from different subjects in a state of eye close and in a state of eye open was significantly distinct due to the strong individual difference of EEG signals. Therefore, different threshold must be confirmed for different users. According to documentation and many experiment results, threshold V_f shall be set as in (1):

$$V_f = V_{open} + 0.8 V_{close} \quad (1)$$

V_{open} is energy of alpha wave in a state of eye open while V_{close} is energy of alpha wave in a state of eye close.

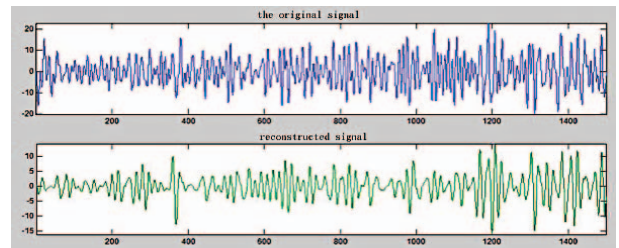
Many experiments results showed that the accuracy and speed of system was ideal when short time eye close which was defined as keep closing eyes for 0.5 second and long-time eye close which was defined as keep closing eyes for 1 second.

This section was used to test whether the subjects had short time eye close movement. The system would start work and implement signal processing when energy of alpha wave in channel O1 exceeded threshold of short time eye close but not reached the threshold of long time eye close. Otherwise, it continued to test to achieve asynchronous BCI.

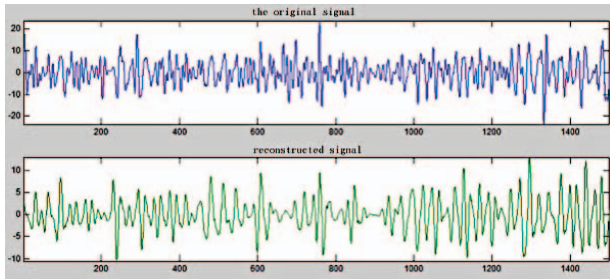
B. Feature Extraction and Pattern Recognition

Motor imagery EEG is a complex spontaneous EEG with the feature of event-related synchronization (ERS) and event-related desynchronization (ERD). Optimal feature vector was get by analyzing this feature and then motor intention could be judged by model identification.

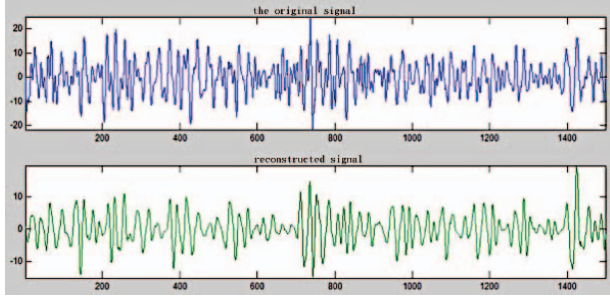
As sampling rate of EEG device was 250 Hz and frequency band width decomposed at the fourth layer was 8 Hz that was approximate to the range of motor imagery EEG, this paper used EEG filtered by Db4 wavelet to carry out the 4-layer decomposition. Frequencies of node (4, 1) and node (4, 3) were 7.81-15.62Hz and 15.62-23.43Hz respectively containing rhythm μ and rhythm β related to motor imagery. So Db4 wavelet was used to decompose the 4-layer motor imagery EEG and signals of node (4, 1) and node (4, 3) were kept. All the rest of the nodes and channels is zero. EEG signal of C4 channel, C3 channel, Cz channel and T4 channel were reconstructed and amplitude of signal, it was less than amplitude of original signal. And then calculation for energy of rebuilt signal every 0.5 seconds was done. Finally, signal reconstruction of corresponding channel to four motor imageries, imagine the ECG upon these movements of left and right hand, feet and tongue, was plotted. And it was shown in Fig. 5.



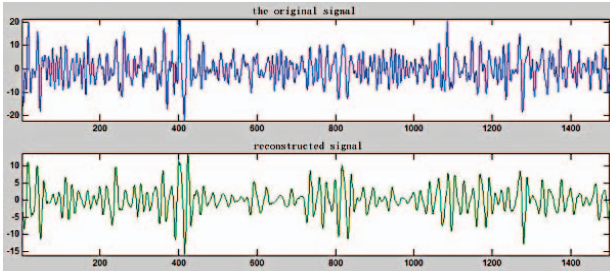
(a)channel C4 of left hand



(b) channel C3 of right hand



(c) channel Cz of feet



(d) channel T4 of tongue

Figure 5 signal reconstruction of corresponding channel to four motor imageries

Signal obtained after wavelet package decomposition and reconstruction contained all the information of rhythm μ and β . It was observed that amplitude of signal decreased much slightly than amplitude of original one and there was an obvious ERD phenomenon in reconstruction.

Pattern Recognition mainly took Support Vector Machine (SVM) classifier to classify four types of motor imagery EEG. One-to-one method was used to design SVM classifier because samples were small and categories were less. Six classifiers for [left hand, right hand], [left hand, tongue], [left hand, feet], [tongue, feet], [tongue, right hand] and [feet, right hand] were designed.

The procedures of classifying motor imagery EEG by SVM classifier were as follows:

1) Label all the training sample and set class labeling. Training sample is as in (2):

$$(x_i, y_i), x_i \in \mathbb{R}^n, y_i \in \{-1, 1\}, i=1, \dots, n; \quad (2)$$

2) Select kernel function and penalty factor. Solve optimal solution (3) by optimization function;

$$\alpha^* = (\alpha_1^*, \dots, \alpha_n^*) \quad (3)$$

3) Solve classifying threshold by formula as in (4):

$$\beta^* = y_i - y_i \alpha^* K(x_i, x_j) \quad (4)$$

4) Solve decision function through taking α^* and β^* into (5):

$$f(x) = \text{sgn} \{ \alpha_i^* y_i K(x_i, x_j + \beta^*) \} \quad (5)$$

5) Obtain classifying number by taking test sample into decision function;

When choose feature extraction and pattern classification algorithms in the practical application, the factors of classification accuracy, complexity, running speed and time consumption need to be considered.

This paper took Support Vector Machine (SVM) classifier to classify the MI vector obtained from wavelet packet decomposition (WPD) and short-time Fourier transform (STFT). Optimal classifying method was selected by comparing results to make preparation for online analysis.

SVM Classifier was used to classify the MI vector obtained from wavelet packet decomposition (WPD) and short-time Fourier transform (STFT). Results were shown in Fig. 6 and Fig. 7.

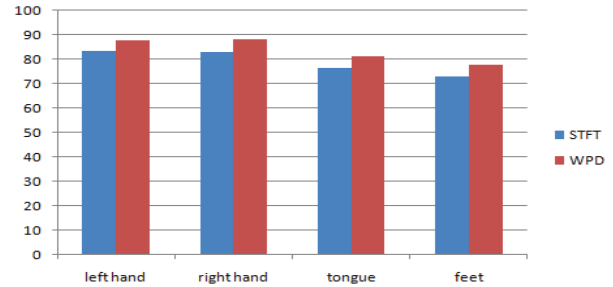


Figure 6 MI classifying accuracy of each channel

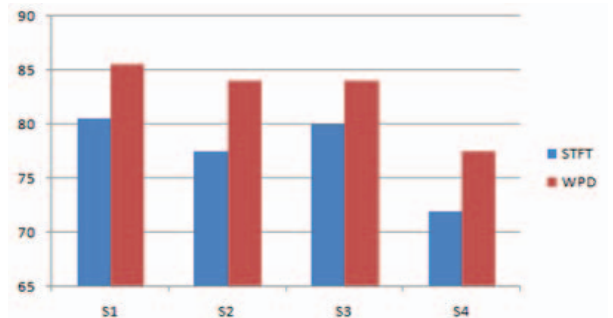


Figure 7 MI classifying accuracy of each subject

The classification results obtained by the two methods about 4 movements was shown in fig. 6, and the classification results obtained by the two methods about 4 subjects was shown in fig. 7.

It suggested that classifying results obtained from wavelet packet decomposition were more optimal than that of short-time Fourier transform. It also showed that wavelet packet decomposition was much more suitable for the MI EEG extraction. Therefore, wavelet packet decomposition was used to extract feature vector in real-time online system.

C. Formal Experiment

There are five healthy students to attend experiments in a relatively quiet laboratory. The seven subjects selected were students in the Lab, aged 23-26. They are healthy and there is no history of mental illness. MI training to subjects was carried out prior to formal experiment to assure the smooth operation of the experiment.

The operability of the system is very important for the operation of the user, how to design and arrange the reasonable operation process, not only in line with the user's usage, but also can make the system run smoothly, and complete the tasks that the user needs. This is the key question we should consider in the design of the system.

The subjects without any hint controlled the cursor movement at up, down, left and right four directions autonomously by EEG and click the confirmation after moving to targeted area of web, which would realize the function of simple website browser without mouse control. Alpha wave produced by long time eye-close controlled to implement mouse click instruction and alpha wave produced by short time eye-close achieved asynchronous control for motor imagery. Theoretically, it can reach ideal location by moving 10 times respectively. Recording actual moving times and completion time as well as calculating the accuracy shall be done.

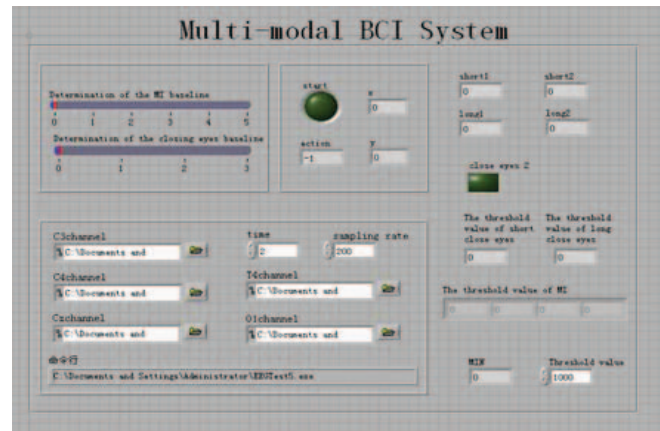


Figure 8 initialization

- 1) The subjects sat well in the armchair with a 65cm distance to computer and install electrodes well to complete preparation.
- 2) Open experiment platform based on LabVIEW and initialization as shown in Fig. 8 ;Collect EEG of subjects and confirm threshold. Then open a browser and move the cursor to initiation location.
- 3) The subjects moved cursor to four appointed locations and click it .
- 4) After experiment, recording actual moving times and completion time as well as calculating the accuracy shall be done.

IV. EXPERIMENT RESULTS AND ANALYSIS

According to the above-mentioned methods, carrying out the statistics of experiment time and accuracy numbers to five movements of 7 subjects was done. The results were shown in Table 1 .

TABLE I. REAL-TIME ONLINE RESULT

Subject	Left hand	Right hand	feet	tongue	Long time eye close	Experiment time/s	Accuracy
S1	12	12	11	11	2	290	86.2%
S2	12	12	11	13	2	297	85.1%
S3	13	12	11	12	2	302	84.0%
S4	—	16	13	14	3	—	—
S5	13	11	11	12	2	315	85.7%
S6	12	12	11	11	2	299	87.5%
S7	11	13	12	11	3	311	84.0%
Accuracy	81.0%	85.2%	89.6%	86.3%	95.2%		

As seen from the results, the accuracy of four types of motor imagery was above 80% and the classification accuracy of research was at a ideal level. It proved that this system is feasible and has a high application

value .Compared to theoretical results, the accuracy in this paper was much more ideal. Hence, the system had certain reliability and algorithm used was available. The Multi-modal BCI system designed in this paper is just a prototype,

still in the laboratory research stage. There are also a lot of work to be further improved. How to improve the stability, accuracy, real time and practicality of BCI system, is still the focus of the next phase of BCI research.

The experiment result for subject 4 was not good. The accuracy was very low although S4 was trained and was able to complete the task of website browser. Assumption was made that S4 is likely to be “EEG blind” and his EEG was very weak. EEG signal has particularity and complexity, different subjects' EEG signals are not the same, even the same subjects will be different due to the different status of the individual and the different external environment . Therefore, adaptable algorithm to signal need to be improved.

V. CONCLUSION

BCI is a new filed developed in recent years and surprises us gradually, which provides a new method to resolve the difficult problems in our life. As for the issues on improvement of systematic usability and accuracy as well as increase of control degrees of freedom, this paper carried out a deep study on EEG process and designed an online BCI system for website browser. After experiment verification, the results suggested that algorithm used in this paper had a good feasibility and the system had a good stability. However, EEG of different subjects are different and even EEG of the same subject also varies because of individual state and exterior environment. Hence, adaptable algorithm to signal is the research focus in the future.

At present, the brain - computer interface is still in the laboratory research level, and in the development stage, there is still a certain distance from the product. How to improve the application of brain computer interface system is an urgent problem to be solved.

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